

Satellite-Based Multi-Indicator Assessment of Urban Lake Restoration: A Case Study of Bommasandra Lake, South Bengaluru

Mahenoor Moka¹, Avni Sodani², Tahir Tamam V², Aiman Fathima Shaikh²,
Y. Suresh² and Azhar Baig¹

¹Department of Computer Science and Data Science,

²Department of Computer Science and Artificial Intelligence,

Ballari Institute of Technology and Management, Ballari, Karnataka, India

mahi090405@gmail.com, a.sodani.2315@gmail.com, tahir.tamam.005@gmail.com,
shaikhaiman250@gmail.com, ysuresh@bitm.edu.in, azharbaig@bitm.edu.in

Abstract

Urban lake restoration requires evidence beyond visual inspection. This study evaluates Bommasandra Lake using satellite-derived NDWI, NDVI, land surface temperature, land-use/land-cover, and terrain indicators over 2019–2025. NDWI increased from -0.183 to $+0.050$, NDVI declined from 0.302 to 0.086 , and reported surface temperature decreased by approximately 2.1°C , indicating a transition toward a more water-dominated and thermally moderated basin.

Keywords— urban lake restoration, remote sensing, NDWI, NDVI, land surface temperature, LULC, DEM, monitoring and verification

I. Introduction

Urban lakes provide water storage, drainage regulation, habitat, and local climate-moderation functions, but rapid urbanisation, sediment accumulation, invasive vegetation, and damaged inflow structures can reduce these services. Conventional restoration evaluation often relies on field observations or isolated before–after photographs. Such approaches may not provide a spatially consistent record of change across the entire restoration boundary.

Satellite remote sensing offers repeatable observations of surface water, vegetation, thermal conditions, and terrain. A combined indicator framework can therefore describe restoration as a multidimensional transition rather than as a single-index change. In this study, Bommasandra Lake in South Bengaluru is assessed using a report-derived monitoring record covering 2019–2025. The analysis combines spectral, thermal, land-cover, and terrain evidence within a validated 7.633-ha basin.

The objective is to determine whether the reported restoration period is associated with measurable changes in water presence, terrestrial vegetation dominance, surface thermal conditions, and basin structure. The study does not treat satellite indices as

direct substitutes for field measurements of groundwater level, biodiversity, dissolved oxygen, or water quality.

II. Related Work

Normalized Difference Vegetation Index (NDVI)[1], is widely used to characterise vegetation activity from red and near-infrared reflectance [1]. Normalized Difference Water Index (NDWI)[2] uses spectral contrast to enhance water-related features and support surface-water mapping [2]. Thermal infrared observations from Landsat have also been used to estimate land surface temperature[3] and investigate urban thermal patterns [3].

Previous remote-sensing studies commonly focus on one indicator, such as water extent or vegetation response. However, an urban lake may show a decrease in NDVI because terrestrial vegetation has been replaced by open water; therefore, interpreting NDVI without NDWI or land-cover context can be misleading. The present work uses a multi-indicator interpretation in which water, vegetation, thermal, terrain, and land-cover observations are read together.

III. Motivation

The Bommasandra report documents a lake basin that changed from a fragmented, vegetation-dominated and water-deficient condition to a more coherent water-dominated condition after restoration activities. A research-oriented assessment is needed to organise these observations into a reproducible framework, distinguish directly measured indicators from ecological interpretation, and demonstrate how a compact set of satellite-derived metrics can support restoration monitoring and verification.

IV. Problem Domain

The problem domain is post-restoration assessment of small urban freshwater systems. Such systems are spatially limited, seasonally variable, and strongly affected by surrounding built-up growth, sedimentation,

vegetation encroachment, rainfall variability, and inflow management. The assessment must therefore be spatially constrained to the lake boundary and interpreted over comparable temporal windows.

V. Problem Definition and Statement

The problem is to assess whether the Bommasandra Lake restoration period is associated with measurable changes in surface-water presence, vegetation composition, thermal behaviour, and basin suitability using satellite and terrain indicators.

The study statement is:

A multi-indicator satellite framework combining NDWI, NDVI, LST, LULC, and DEM-derived terrain descriptors can provide a spatially consistent evidence base for evaluating the direction and magnitude of urban lake restoration change.

VI. Innovative Content

The contribution of this work is the integration of five complementary evidence dimensions within one lake-scale assessment:

1. NDWI for surface-water condition;
2. NDVI for terrestrial and marginal vegetation response;
3. Landsat thermal information for surface-temperature change[4];
4. LULC for the transition between terrestrial and aquatic surface classes; and
5. SRTM-derived elevation and slope for interpreting basin form and water-retention context.

The key interpretive contribution is that a negative NDVI change is not automatically classified as ecological degradation. In a restored lake basin, reduced NDVI may indicate the displacement of terrestrialised vegetation by open water when supported by a simultaneous positive NDWI change and LULC transition.

VII. Problem Formulation, Representation and Design

A. Study Area

Bommasandra Lake is located in the Bommasandra–Attibele urban-peripheral corridor of South Bengaluru, Karnataka, India. The report defines a 7.633-ha assessment boundary with a reported perimeter of 1,345.80 m and centroid near 12.8087°N, 77.6898°E. The basin has a reported elevation range of 885–894 m, mean elevation of 888.82 m, mean slope of 2.58°, and maximum slope of 7.05°.

B. Temporal Design

The report uses 2019–2025 as the overall monitoring period. The pre-restoration reference is described as December 2019–December 2020, while the post-restoration reference is described as December 2024–December 2025. The comparison is therefore treated



Fig. 1: Aerial view of Bommasandra Lake and its surrounding urban landscape.

as a baseline-versus-current assessment rather than as a controlled experiment.

C. Data Sources

The report identifies Landsat imagery for spectral and thermal analysis, SRTM data for terrain modelling[5], and ESA land-cover information for land-use/land-cover interpretation[6]. All reported indicators are spatially constrained to the defined lake boundary.

D. Indicator Formulation

NDVI is computed as[1]

$$NDVI = \frac{NIR - Red}{NIR + Red}. \quad (1)$$

NDWI is computed as[2]

$$NDWI = \frac{Green - NIR}{Green + NIR}. \quad (2)$$

For a baseline value I_b and post-restoration value I_p , the absolute change is

$$\Delta I = I_p - I_b. \quad (3)$$

For positive baseline values, the relative percentage change is

$$\% \Delta I = \frac{I_p - I_b}{I_b} \times 100. \quad (4)$$

Because NDVI and NDWI may be negative or near zero, the study emphasises absolute change and directional interpretation rather than applying percentage change indiscriminately.

VIII. Solution Methodologies and Problem Solving

The assessment workflow consists of: (i) defining the lake boundary; (ii) selecting comparable pre- and post-restoration observation periods; (iii) processing Landsat spectral and thermal information; (iv) calculating NDVI and NDWI; (v) deriving LST-related summaries; (vi) extracting elevation and slope from

Table 1: Reported baseline and post-restoration indicators

Indicator	Baseline	Current	Change
NDWI	−0.183	+0.050	+0.233
NDVI	0.302	0.086	−0.216
NDVI relative change	–	–	−71.5%
LST	Elevated	Reduced	≈ −2.1°C
Basin area	7.633 ha	7.633 ha	Stable
Mean elevation	888.82 m		Stable
Mean slope	2.58°		Stable

SRTM; (vii) comparing LULC conditions; and (viii) interpreting the indicators jointly. Google Earth Engine was used for satellite-data processing, image filtering, and calculation of environmental indicators [7].

The report describes four restoration activity groups: invasive-vegetation removal and desilting; inlet-channel rehabilitation and bund reinforcement; peripheral vegetation planting and shoreline stabilisation; and continued satellite-based monitoring. These activities are treated as contextual information for interpreting the observed transition, not as independently tested causal variables.

IX. Results and Sensitivity Analysis

A. Spectral Indicator Change

The reported NDWI increased from −0.183 during the baseline period to +0.050 during the current period. The absolute improvement was therefore +0.233. The sign change indicates that the basin’s average spectral response moved from a water-deficient condition toward a water-dominated response under the report’s classification and processing approach.

NDVI decreased from 0.302 to 0.086, an absolute change of −0.216. The report gives this as a 71.5% reduction[8]. In this context, the decline is interpreted together with the NDWI increase and the reported LULC transition: terrestrial or semi-terrestrial vegetation occupying the lakebed was displaced by open water, while some marginal vegetation remained.

B. Thermal Response

The report describes a reduction of approximately 2.1°C in the thermal indicator between the pre-restoration and current conditions. The baseline is characterised by elevated thermal response associated with exposed dry sediment, whereas the current condition is described as thermally reduced with increased water coverage. The result supports a change in surface thermal behaviour, but it should not be interpreted as a direct measurement of near-surface air temperature or community-scale climate change.

C. Terrain and Land-Cover Context

The SRTM-derived elevation range of 885–894 m and mean slope of 2.58° indicate a gently sloping basin. These terrain descriptors are stable structural properties and provide context for water retention and drainage interpretation. The report also describes a land-cover transition from terrestrial encroachment in the baseline period to open water as the dominant current classification.

D. Sensitivity and Uncertainty

The assessment is sensitive to seasonal water level, rainfall timing, image acquisition date, cloud and shadow screening, sensor resolution, index formulation, and the spatial boundary used for aggregation. In addition, a single mean index value can conceal within-basin heterogeneity. The report’s thermal discussion contains two temporal descriptions: the summary refers to 2019–2020 versus 2024–2025, while the LST section mentions 2019–2020 versus 2023–2024. This must be reconciled against the original processing records before final submission[9].

X. Data Model

The conceptual data model contains five layers:

1. **Boundary layer:** lake polygon, area, perimeter, and centroid;
2. **Spectral layer:** red, green, and near-infrared observations, NDVI, and NDWI;
3. **Thermal layer:** Landsat thermal observations and LST summaries;
4. **Terrain layer:** elevation and slope derived from SRTM; and
5. **Classification layer:** baseline and current LULC classes and restoration-suitability interpretation.

Each layer is clipped or summarised within the same lake boundary so that the indicators remain spatially comparable.

XI. Comparison of Results

The multi-indicator comparison shows a consistent directional pattern across the assessed environmental variables. The mean NDWI increased from −0.183 during the baseline period to 0.050 during the post-restoration period, representing an absolute increase of 0.233. This indicates a stronger open-water spectral response within the assessment boundary. In contrast, the mean NDVI decreased from 0.302 to 0.086, suggesting a shift from a vegetation-dominated or terrestrialised surface signature toward an open-water-dominated surface signature.

The LULC comparison provides a categorical interpretation of this transition by indicating an increase in the spatial dominance of open water within the lake boundary. The reported land surface temperature also decreased by approximately 2.1°C, indi-

cating an improvement in the thermal condition during the compared periods. However, the magnitude of LST change may be influenced by seasonal conditions, atmospheric effects, acquisition dates, and image-processing procedures. The spatial comparison of land surface temperature between the baseline and post-restoration periods is shown in Fig. 2.



Fig. 2: Comparison of land surface temperature (LST) between the baseline period and the post-restoration period. The baseline image shows elevated thermal conditions associated with dry exposed sediment, whereas the post-restoration image shows a reduced surface-temperature pattern associated with increased water coverage.

Terrain variables remained stable because they describe the underlying basin rather than restoration performance. The DEM analysis reported an elevation range of 885–894 m, a mean elevation of 888.82 m, a mean slope of 2.58°, and a maximum slope of 7.05°. These values provide physical context for understanding the lake’s depression, drainage characteristics, and potential water accumulation zones.

The indicators therefore provide complementary evidence. NDWI describes the spectral response associated with surface water, NDVI describes vegetation dominance, LULC describes categorical surface transition, LST describes thermal response, and DEM-derived variables describe the physical context of the basin. Considering these indicators together provides a more comprehensive assessment than relying on a single index.

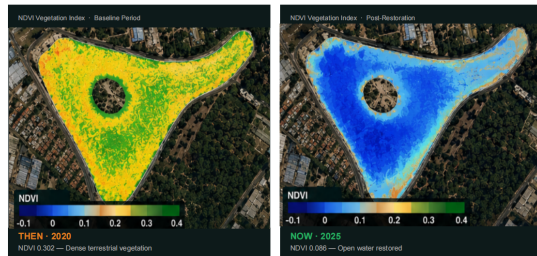


Fig. 3: Comparison of the Normalized Difference Vegetation Index (NDVI) between the baseline period and the post-restoration period. The mean NDVI decreased from 0.302 in 2020 to 0.086 in 2025, indicating a shift from a vegetation-dominated surface signature toward an open-water-dominated surface signature.

XII. Justification of Results

The reported NDWI increase is consistent with the visual description of a fragmented basin becoming more coherent and water-dominated. A shift from a negative mean NDWI value to a positive value indicates a stronger water-related spectral response in the post-restoration period. Nevertheless, NDWI is an indirect indicator and may also be affected by shallow water, wet soil, aquatic vegetation, suspended material, and atmospheric conditions.

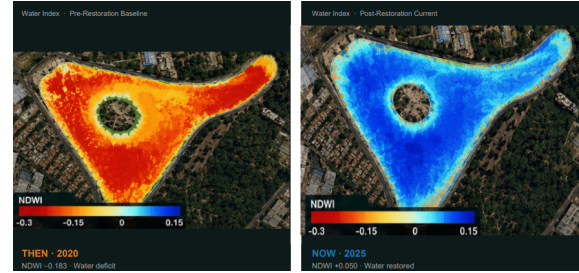


Fig. 4: Comparison of the Normalized Difference Water Index (NDWI) between the pre-restoration baseline and post-restoration periods. The mean NDWI increased from -0.183 in 2020 to $+0.050$ in 2025, indicating a stronger water-related spectral response within the assessment boundary.

The simultaneous NDVI decline is consistent with the replacement of terrestrialised lakebed vegetation by open water, rather than necessarily indicating a loss of ecological value. The baseline NDVI value of 0.302 suggests a stronger vegetation-related spectral response, whereas the post-restoration value of 0.086 is more consistent with the increased presence of open water. Therefore, the NDVI reduction should be interpreted in relation to the NDWI and LULC results instead of being treated independently as negative vegetation performance.

The reported thermal reduction is physically plausible because water generally has greater heat capacity than exposed dry sediment and can moderate surface temperature. However, the observed LST difference may also depend on the acquisition season, time of satellite overpass, atmospheric conditions, cloud masking, and the processing method used to derive land surface temperature. Thus, the reported reduction should be interpreted as a satellite-observed thermal change rather than a direct measurement of long-term climatic improvement.

The terrain results support the interpretation that the basin has a defined depression and gentle average slope. Such characteristics may influence surface runoff, water retention, and drainage behaviour. However, slope and elevation alone cannot prove groundwater recharge. Direct recharge, biodiversity recovery, dissolved oxygen, water quality, or flood attenuation would require field measurements, hydrological

modelling, or independent observations.

Accordingly, the results demonstrate measurable changes in satellite-derived surface indicators within the assessment boundary. They provide evidence of increased water-related spectral response, reduced vegetation dominance over the lake surface, a reported thermal change, and a stable physical terrain context. These findings can support restoration monitoring and decision-making, while potential ecosystem services are described as possible implications rather than directly quantified outcomes.

XIII. Implications for Restoration Monitoring

The results demonstrate how satellite-derived indicators can be organised into a repeatable monitoring framework for urban lake restoration. NDWI can be used to track changes in the spatial and spectral presence of surface water across successive observation dates. A positive change in mean NDWI may indicate increased water coverage or improved water-related spectral response, although the result should be interpreted together with acquisition conditions and the selected lake boundary.

NDVI provides a complementary measure of vegetation-related surface response. In a terrestrialised or partially dry lake basin, relatively high NDVI values may be associated with terrestrial vegetation growing over exposed lakebed areas. Therefore, a reduction in NDVI does not necessarily represent ecological degradation when it occurs simultaneously with an increase in NDWI and a transition toward open water. This highlights the importance of interpreting vegetation and water indices jointly rather than using a single indicator as the sole measure of restoration success.

LULC classification can support the conversion of continuous spectral observations into interpretable surface categories such as open water, vegetation, bare soil, and built-up area. Repeated classification can help identify whether changes are concentrated within the lake boundary or occur in the surrounding urban landscape. However, classification accuracy depends on training data, class separability, image quality, and the classification method.

LST can provide additional information about the thermal condition of the lake and its surroundings. A reduction in surface temperature may be associated with increased water coverage, changes in exposed sediment, or surrounding land-cover changes. Since LST is sensitive to season, atmospheric conditions, and satellite acquisition time, future monitoring should compare images acquired under similar seasonal and atmospheric conditions.

The combined use of NDWI, NDVI, LULC, LST, and terrain variables can therefore support a struc-

tured restoration-monitoring workflow. The framework can be repeated at monthly, seasonal, or quarterly intervals to identify the direction and persistence of environmental changes. Such monitoring can assist restoration agencies in identifying areas requiring further inspection and in maintaining a transparent record of satellite-observed changes.

The terrain results support the interpretation that the basin has a defined depression and gentle average slope. However, slope alone cannot prove groundwater recharge. Direct recharge, biodiversity recovery, dissolved oxygen, or flood attenuation would require field measurements, hydrological modelling, or independent observations. Accordingly, these outcomes are described here as potential ecosystem services supported by the restored water presence, not as directly quantified results.

XIV. Limitations

This study is based primarily on satellite-derived indicators and the information available in the Bommasandra Lake restoration assessment. The analysis therefore describes changes in surface characteristics rather than providing a complete ecological evaluation. The reported NDWI, NDVI, and LST values are period-level summaries, and acquisition-date-level information is required to fully evaluate seasonal and atmospheric influences.

The study does not directly measure groundwater recharge, dissolved oxygen, biodiversity, water quality, sediment depth, or flood attenuation. These parameters require field observations, hydrological modelling, or independent monitoring data. Similarly, the reported LULC transition depends on the classification procedure and should be supported by accuracy assessment and validation samples.

The reported LST reduction should also be interpreted cautiously because the compared periods may differ in acquisition season, atmospheric condition, and processing method. Future work should include uncertainty estimates, consistent image-selection criteria, independent validation, and comparison with a nearby non-restored reference water body. These steps would improve the reliability and generalisability of the proposed monitoring framework.

XV. Conclusion

This study organised the Bommasandra Lake restoration report into a multi-indicator satellite assessment. Across the reported 2019–2025 period, NDWI increased from -0.183 to $+0.050$, NDVI decreased from 0.302 to 0.086 , the reported land-cover condition shifted toward open water, and surface thermal conditions decreased by approximately 2.1°C . The combined evidence indicates a transition from a terrestrialised, water-deficient basin toward a more coherent

water-dominated state.

The main value of the framework is its interpretive complementarity: no single index is used in isolation. The study also demonstrates the importance of separating satellite-observed change from ecological claims that require field validation.

XVI. Future Work

Future work should include: (i) acquisition-date-level time series rather than only period averages; (ii) explicit cloud, shadow, and seasonal matching details; (iii) water-area estimation from a stated threshold and validation samples; (iv) independent verification of LST calculations; (v) field measurements of water quality, groundwater level, and biodiversity; (vi) uncertainty bounds and sensitivity tests for boundary and threshold choices; and (vii) comparison with a nearby non-restored reference water body where feasible.

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